

Limit Cycles and Self-Sustained Oscillation

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Recall: Periodic solution and closed orbit

A nonconstant solution $z : \mathbb{R} \rightarrow \mathbb{R}^2$ is **periodic** if there is a number $T > 0$ such that

$$z(t + T) = z(t)$$

for every $t \in \mathbb{R}$. Its least positive period is its **period**. Its image

$$\Gamma = \{z(t) : 0 \leq t \leq T\}$$

is its **closed orbit**.

Definition: Limit cycle

A closed orbit Γ is a **limit cycle** if it is isolated: there is a neighborhood of Γ containing no other closed orbit.

Notation: Distance from an orbit

For $z \in \mathbb{R}^2$ and a compact orbit $\Gamma \subset \mathbb{R}^2$, write

$$\text{dist}(z, \Gamma) = \inf_{p \in \Gamma} \|z - p\|.$$

Definition: Orbital Lyapunov stability

A limit cycle Γ of $\dot{z} = F(z)$ is **orbitally Lyapunov stable** if, for every $\varepsilon > 0$, there is a $\delta > 0$ such that every solution $z(t)$ with

$$\text{dist}(z(0), \Gamma) < \delta$$

is defined for every $t \geq 0$ and satisfies

$$\text{dist}(z(t), \Gamma) < \varepsilon \quad \text{for all } t \geq 0.$$

Definition: Attracting limit cycle

A limit cycle Γ of $\dot{z} = F(z)$ is **attracting** if there is a $\delta > 0$ such that every solution $z(t)$ with

$$\text{dist}(z(0), \Gamma) < \delta$$

is defined for every $t \geq 0$ and satisfies

$$\text{dist}(z(t), \Gamma) \rightarrow 0 \quad (t \rightarrow \infty).$$

Definition: Asymptotic orbital stability

A limit cycle is **asymptotically orbitally stable** if it is both orbitally Lyapunov stable and attracting.

Definition: Half-stable limit cycle

A planar limit cycle Γ separates the plane into an inside and an outside. It is **half-stable** if it attracts every sufficiently close trajectory on one side, whereas every sufficiently close trajectory on the other side eventually leaves some fixed neighborhood of Γ .

Notation: Stable limit cycle

In these notes, **stable limit cycle** means asymptotically orbitally stable limit cycle.

Visualization: Closed families versus isolated cycles

phase portrait	isolated?	amplitude after perturbation
linear center	no	new amplitude persists
undamped pendulum librations	no	new energy persists
attracting limit cycle	yes	returns to selected orbit

Proposition: Radial construction principle

Let $I \subseteq (0, \infty)$ be an open interval, let $R : I \rightarrow \mathbb{R}$ and $\Omega : I \times S^1 \rightarrow \mathbb{R}$ be locally Lipschitz, and consider

$$\dot{r} = R(r), \quad \dot{\theta} = \Omega(r, \theta), \quad (r, \theta) \in I \times S^1.$$

Suppose $a \in I$ is an isolated zero of R and

$$\Omega(a, \theta) \neq 0$$

for every $\theta \in S^1$. Then $r = a$ is a limit cycle.

Proof. Since $R(a) = 0$, uniqueness makes $r = a$ invariant. The continuous nonzero function $\Omega(a, \theta)$ has one sign on the connected circle, so the solution traverses $r = a$ periodically with period

$$T = \left| \int_0^{2\pi} \frac{d\theta}{\Omega(a, \theta)} \right|.$$

On any closed orbit, $r(t)$ is periodic. The scalar autonomous equation $\dot{r} = R(r)$ has no nonconstant periodic solution, so $r(t)$ must be a constant zero of R . Isolation of the zero a therefore excludes every nearby closed orbit, including noncircular ones. $\square$

Corollary: Radial stability test

Let $I \subseteq (0, \infty)$ be open, let $R : I \rightarrow \mathbb{R}$ and $\Omega : I \times S^1 \rightarrow \mathbb{R}$ be locally Lipschitz, and consider

$$\dot{r} = R(r), \quad \dot{\theta} = \Omega(r, \theta).$$

Suppose $a \in I$ is an isolated zero of R and $\Omega(a, \theta) \neq 0$ for every $\theta \in S^1$. Then $r = a$ is a limit cycle with the following classification, according to the signs of $R(r)$ for r sufficiently close to a :

$R(r)$ for $r < a$	$R(r)$ for $r > a$	type
$R(r) > 0$	$R(r) < 0$	stable
$R(r) < 0$	$R(r) > 0$	not orbitally Lyapunov stable
$R(r) > 0$	$R(r) > 0$	half-stable: attracts from inside
$R(r) < 0$	$R(r) < 0$	half-stable: attracts from outside

Proof. The distance to the circle $r = a$ is $|r - a|$. Apply the scalar phase-line test to $\dot{r} = R(r)$: motion toward a decreases this distance to zero; motion away from a eventually leaves a fixed neighboring annulus. These are precisely the stated orbital classifications. $\square$

Question: The canonical attracting limit cycle

Analyze

$$\dot{r} = r(1 - r^2), \quad \dot{\theta} = 1.$$

Interpret these as polar equations in the plane, so all coordinate pairs $(0, \theta)$ represent the single Cartesian origin. Find and classify every nonnegative invariant radius.

Solution. The radial equilibria are $r = 0$ and $r = 1$. The sign chart is

r	$(0, 1)$	1	$(1, \infty)$
$\dot{r}$	+	0	-

so the origin repels nearby nonzero states and $r = 1$ is an attracting limit cycle with period $T = 2\pi$.

Question: Solve the canonical radial equation

For $t \geq 0$, solve

$$\dot{r} = r(1 - r^2), \quad r(0) = r_0 > 0.$$

Solution. Set $q = r^2$. Then

$$\dot{q} = 2q(1 - q).$$

For $u = 1/q$,

$$\dot{u} = -\frac{\dot{q}}{q^2} = 2(1 - u),$$

and hence

$$u(t) = 1 + (r_0^{-2} - 1)e^{-2t}.$$

Since $r(t) > 0$,

$$r(t) = \frac{1}{\sqrt{1 + (r_0^{-2} - 1)e^{-2t}}} \rightarrow 1 \quad (t \rightarrow \infty).$$

Question: What Cartesian system has the canonical cycle?

Convert

$$\dot{r} = r(1 - r^2), \quad \dot{\theta} = 1$$

to Cartesian coordinates.

Solution. With $x = r \cos \theta$ and $y = r \sin \theta$, we have

$$\dot{x} = \dot{r} \cos \theta - r \dot{\theta} \sin \theta, \quad \dot{y} = \dot{r} \sin \theta + r \dot{\theta} \cos \theta.$$

Since $r^2 = x^2 + y^2$,

$$\begin{aligned} \dot{x} &= x(1 - x^2 - y^2) - y, \\ \dot{y} &= y(1 - x^2 - y^2) + x. \end{aligned}$$

Conversely,

$$r\dot{r} = x\dot{x} + y\dot{y}, \quad r^2\dot{\theta} = x\dot{y} - y\dot{x},$$

which recover the radial and angular equations for $r > 0$.

Question: Two limit cycles

Classify the positive invariant radii of

$$\dot{r} = r(r^2 - 1)(4 - r^2), \quad \dot{\theta} = 1.$$

Solution.

r	(0, 1)	1	(1, 2)	2	(2, ∞)
$\dot{r}$	-	0	+	0	-

so the cycle $r = 1$ repels nearby radii on both sides and is not orbitally Lyapunov stable, whereas $r = 2$ is a stable limit cycle. The origin attracts every state with $0 < r < 1$.

Question: A half-stable limit cycle

Classify the positive invariant radii of

$$\dot{r} = r(r^2 - 1)^2(4 - r^2), \quad \dot{\theta} = 1.$$

Solution.

r	(0, 1)	1	(1, 2)	2	(2, ∞)
$\dot{r}$	+	0	+	0	-

so $r = 1$ attracts from the inside and repels from the outside: it is half-stable. The cycle $r = 2$ remains stable.

Theorem: Homogeneous linear systems have no limit cycles

No planar homogeneous linear system

$$\dot{z} = Az$$

has a limit cycle.

Proof. Suppose $z(t)$ is a nonconstant periodic solution. For every $c > 0$,

$$\frac{d}{dt}(cz(t)) = A(cz(t)),$$

so $cz(t)$ is another nonconstant periodic solution. Its maximum distance from the origin is c times that of the original orbit, so the two orbits are distinct when $c \neq 1$. Taking c arbitrarily close to 1 produces closed orbits arbitrarily close to the original one. Thus no closed orbit is isolated. $\square$

Question: What does the van der Pol energy balance prove?

For

$$\ddot{x} + \mu(x^2 - 1)\dot{x} + x = 0, \quad \mu > 0,$$

compute the derivative of the linear-oscillator energy and state its exact consequence.

Solution. Let

$$v = \dot{x}, \quad E(x, v) = \frac{1}{2}v^2 + \frac{1}{2}x^2.$$

Then

$$\begin{aligned} \dot{E} &= v(\dot{v} + x) \\ &= -\mu(x^2 - 1)v^2. \end{aligned}$$

Hence

$$\begin{array}{l|l} |x| < 1, v \neq 0 & \dot{E} > 0 \\ |x| > 1, v \neq 0 & \dot{E} < 0 \\ |x| = 1 \text{ or } v = 0 & \dot{E} = 0 \end{array}$$

Energy is injected exactly when $|x| < 1$ and $v \neq 0$, dissipated exactly when $|x| > 1$ and $v \neq 0$, and instantaneously unchanged when $|x| = 1$ or $v = 0$. This pointwise sign change supplies an amplitude-selection mechanism, but the calculation alone does not prove existence, uniqueness, or stability of a periodic orbit.

Idea: Self-sustained oscillation

In a radial or energy-balance description of a self-sustained oscillator,

small amplitude $\longrightarrow$ net outward motion,

large amplitude $\longrightarrow$ net inward motion,

suggests that the dynamics may select an intermediate amplitude. This is a heuristic: it does not by itself prove that a periodic orbit exists or that such an orbit is isolated, unique, or stable. Each of those conclusions requires additional arguments.